PYTHON

If I Had a Python Interview in 48 Hours, These 8 Topics Only

The 8 Python topics data analyst interviews actually test, with a 48-hour schedule, the key code for each and the questions to expect.

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The 48-hour schedule

Two days, 4 topics each. Code every example yourself.

- **Day 1 morning**

  Topic 1: core data types. Topic 2: loops, comprehensions, functions.

- **Day 1 evening**

  Topic 3: loading and inspecting data. Topic 4: cleaning.

- **Day 2 morning**

  Topic 5: filtering and groupby. Topic 6: merge and pivot.

- **Day 2 evening**

  Topic 7: dates. Topic 8: charts. Then one timed mock task.

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Topic 1: Core data types

Asked as warm-up questions in almost every round.

List vs tuple vs set vs dict

List: ordered, changeable. Tuple: ordered, fixed. Set: unique values. Dict: key-value pairs.

Typical question

"Remove duplicates from a list but keep the order."

```
seen = set()
result = [x for x in items
          if not (x in seen or seen.add(x))]
# or simply: list(dict.fromkeys(items))
```

Count frequency

Use a dict or Counter.

```
from collections import Counter
Counter(["a","b","a"])   # {"a": 2, "b": 1}
```

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Topic 2: Loops, comprehensions, functions

They check if you can write clean, simple logic.

List comprehension

One-line transform and filter.

```
high = [x * 1.18 for x in prices if x > 1000]
```

Function with default argument

Write small reusable functions.

```
def growth(curr, prev=0):
    if prev == 0:
        return None
    return round((curr - prev) / prev * 100, 1)
```

Typical question

"Find the second largest number in a list without sorting."

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Topic 3 and 4: Load, inspect, clean

The first 10 minutes of any take-home task.

Load and inspect

Always check shape, types and missing values first.

```
df = pd.read_csv("data.csv")
df.shape; df.info(); df.isna().sum()
```

Clean

Duplicates, missing values, types, text.

```
df = df.drop_duplicates("order_id")
df["amount"] = pd.to_numeric(df["amount"],
                             errors="coerce")
df["city"] = df["city"].str.strip().str.title()
```

Typical question

"How do you handle missing values?" Answer: understand why they are missing, then drop, fill or flag. Never fill blindly.

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Topic 5 and 6: Analyse

Groupby, merge and pivot answer most take-home questions.

Filter and groupby

Revenue and order count by region.

```
df[df["amount"] > 0].groupby("region").agg(
    revenue=("amount", "sum"),
    orders=("order_id", "nunique"))
```

Top N per group

Top 3 products by revenue per category.

```
(df.groupby(["category", "product"])["amount"]
  .sum().reset_index()
  .sort_values("amount", ascending=False)
  .groupby("category").head(3))
```

Merge and pivot

Join tables, then make a report view.

```
m = orders.merge(customers, on="cust_id",
how="left")
m.pivot_table(index="region", columns="month",
              values="amount", aggfunc="sum")
```

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Topic 7 and 8: Dates and charts

Almost every dataset has dates. Most tasks end with one chart.

Dates

Convert, then extract parts or resample.

```
df["date"] = pd.to_datetime(df["date"])
df["month"] = df["date"].dt.to_period("M")
monthly = df.set_index("date")
["amount"].resample("M").sum()
```

Month-on-month growth

pct_change gives growth from the previous row.

```
monthly.pct_change().mul(100).round(1)
```

One clear chart

Bar for comparison, line for trend. Always add a title.

```
monthly.plot(kind="line", title="Monthly
revenue")
```

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Questions to expect

Practise answering these out loud in under a minute each.

- **loc vs iloc**
  loc uses labels, iloc uses positions.
- **merge vs concat**
  merge joins on keys like SQL. concat stacks tables.
- **apply vs vectorised**
  Vectorised column operations are much faster. Use apply only when there is no built-in way.
- **Large file that does not fit in memory**
  Read in chunks, load only needed columns with usecols, set smaller dtypes.
- **Walk me through your analysis**
  Question, cleaning steps, method, finding, recommendation.

Last step on day 2: take any sales CSV and answer 5 questions in 45 minutes. That is what most take-home rounds look like.

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